Sahaj Bhadja

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Summary

Backend engineer with 2+ years building production APIs, LLM orchestration, document pipelines, and scalable data services. Experienced in Python, FastAPI, Flask, SQL, PostgreSQL, AWS, Docker, distributed workflows, semantic search, testing, monitoring, and reliable deployments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python APIs** and Azure SQL services for 500+ users, optimizing indexed queries, schemas, and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Restructured a fault-tolerant **LLM orchestration system** with explicit job states and failure isolation, processing 1,000+ AI jobs while tripling throughput and improving observability, recovery, and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Flask REST services** reliably handling 10,000+ daily requests, implementing secure document ingestion, authentication, grading workflows, and error handling while debugging production issues affecting 150+ users.
- Created multithreaded document processing for 1,000+ files with validation, **AWS RDS** persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable **Python and PyTorch pipelines** processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured evaluation, performance monitoring, validation checks, and systematic debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services and data integrations** using AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and data-processing efficiency 68% across applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect investigation, release verification, and reliable deployments across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular **Node.js services** across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, securely processing 2 million+ Azure SQL records through documented software interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Skills

Languages & Backend: Python, Java, JavaScript, TypeScript, SQL, Flask, FastAPI, Node.js, REST APIs, microservices, object-oriented programming, multithreading, concurrency
LLM & Retrieval: LLM APIs, prompt pipelines, semantic search, vector search, FAISS, Pinecone, embeddings, document ingestion, metadata processing, model evaluation, PyTorch, AI-assisted workflows
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, AWS, Azure, GCP, S3, AWS Glue, relational databases, schema design, query optimization, caching
Engineering & Operations: Docker, Kubernetes, Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, debugging, monitoring, logging, failure recovery, performance analysis, documentation, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Built a **semantic document retrieval system** over 50,000+ documents using transformer embeddings and FAISS, enabling contextual question answering and summarization with sub-second retrieval through optimized indexing pipelines.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an **AI-powered data extraction workspace** supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable backend supporting 50,000+ users and one million transactions, integrating **secure APIs**, real-time WebSockets, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.